



National University of Computer & Emerging Sciences, Karachi

Department of Computer Science

Final Exam, Summer-2022

Saturday, 13th August 2022, 9:30 am – 12:30 pm



Course Code: MT-1004	Subject: Linear Algebra
Instructor Name: Mr. Muhammad Jamil,	
Student Roll No:	Section:

Instructions:

- Solve the entire questions on answer sheet and return the question paper.
- Read each question completely before answering it. There are 6 questions and 2 pages.
- All the answers must be solved according to the sequence given in the question paper.
- Scientific calculator is allowed and shows all the necessary steps.

Time: 180 minutes

Max Marks: 70 points

----- [5+5+5]

Q-1 a) Write the general solution of linear system whose augmented matrix has been reduced to echelon form

$$[A:b] \sim \begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

Discuss pivot column, free and basic variables, determine the given system is consistent

b) Consider $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x_1, x_2, x_3) = (x_1, x_3, x_1 + x_2)$

Write the standard matrix and show that T is linear matrix transformation

c) Find the basis for null space of the matrix A , what is $\text{rank}[A]$

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

----- [7+3]

Q-2 Consider $A = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$ and characteristic equation of A is $P(\lambda) = \lambda^3 - 6\lambda^2 + 9\lambda$:

- Use Gram-schmidt process to construct an orthogonal matrix P from given eigen vectors
- Find the diagonal matrix D such that $PAP^T = D$

-----[5+5]

Q-3 Evaluate (any two)

- I. Find cosine angle between the vectors with relative to the standard inner product on P_2 and confirm Cauchy-Schwarz inequality hold.

$$\mathbf{p} = -1 + 2x + x^2 \text{ and } \mathbf{q} = 2 - 4x^2$$

- II. Let $U = \begin{bmatrix} 3 & -2 \\ 4 & 8 \end{bmatrix}$, $V = \begin{bmatrix} -1 & 3 \\ 1 & 1 \end{bmatrix}$, Compute $||U||$ & $d(U, V)$

and check for orthogonality of matrix relative to the standard inner product on M_{22}

- III. Confirm that the Cauchy-Schwarz inequality hold for the following vectors

$$\mathbf{u} = (1, 0, 3), \mathbf{v} = (2, 1, -1) \text{ using inner product } \langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + 3u_2v_2 + u_3v_3 \text{ in } R^3.$$

-----[10]

Q-4 Find an orthogonal change of variable that eliminate the cross product term in quadratic for Q and express Q in term of new variables .

$$Q = 5x_1^2 + 2x_2^2 + 4x_3^2 + 4x_1x_2$$

-----[5+5]

Q-5 a) Consider the basis vectors $u_1 = (1, 1, 1)$, $u_2 = (0, 1, 1)$, $u_3 = (0, 0, 1)$

Determine an orthogonal basis $\{v_1, v_2, v_3\}$ and orthonormal basis $\{q_1, q_2, q_3\}$ vectors.

b) Express the matrix $A = QR$ decomposition, where Q matrix obtained by applying the Gram-schmidt process to the column vector of A

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}, \quad Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \end{bmatrix}$$

-----[5+10]

Q-6 a) Find the LU decomposition (use dollittle method) and then solve the system $AX=b$

$$\begin{pmatrix} 1 & 1 & 1 \\ 3 & 1 & 2 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}$$

b) Find the singular value decomposition of $A = \begin{bmatrix} 4 & 6 \\ 0 & 4 \end{bmatrix} = U \Sigma V^T$, Show all the steps.